

Architecture

System Overview



Syntax error in text
mermaid version 10.9.6

Kafka Event Flow (Order Saga)



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Key design decision: Payment only fires AFTER inventory confirms stock is reserved. If inventory fails → order is automatically cancelled → no payment taken.

Data Model (DB per service)

Each service owns its own PostgreSQL database. No cross-service DB queries.

```
userdb      → users (id, email, password_hash, role, active)
productdb   → products (id, name, price, category, stock=inventory)
orderdb     → orders + order_items
inventorydb → inventory (product_id, quantity, reserved, version)
paymentdb   → payments (order_id, amount, status, transaction_id)
```

Redis Usage (3 purposes)

Purpose	Service	Key pattern	TTL
JWT sessions	user-service	<code>session:</code>	24h
Product cache	product-service	<code>product:</code>	15 min
Distributed lock	inventory-service	<code>inventory:lock:</code>	30s

Security Layers

```
Internet
↓
WAF (OWASP rules + rate limit 2000 req/5min + SQLi blocking)
↓
ALB (SSL/TLS termination — HTTPS only)
↓
Kong (rate limiting 1000/min + correlation IDs + JWT validation)
↓
Services (Spring Security + JWT + input validation)
↓
Databases (separate credentials per service, no cross-service access)
```

Floci vs Real AWS

Component	Floci (local)	Real AWS
VPC / Subnets / SGs	✅ Full API simulation	AWS EC2
ALB + listeners	✅ Port 80 redirect works; port 443 TLS not implemented	AWS ELB
WAF	✅ API works	AWS WAF
ACM (SSL cert)	✅ API works	AWS ACM
Route 53	✅ API works	AWS Route 53
ECR (push/pull)	✅ Real Docker Registry v2 at localhost:5100	AWS ECR
EKS (cluster)	✅ Real k3s cluster provisions	AWS EKS
SES (email)	✅ Real API, emails logged	AWS SES

S3	 Full API	AWS S3
CloudWatch Logs	 API works	AWS CloudWatch